

FIT TRACKER



*Application
Discussion*

Final Presentation

Presented by: Amer Amer, Nada Belafqih, Ronald Berle, Nairi Keeney



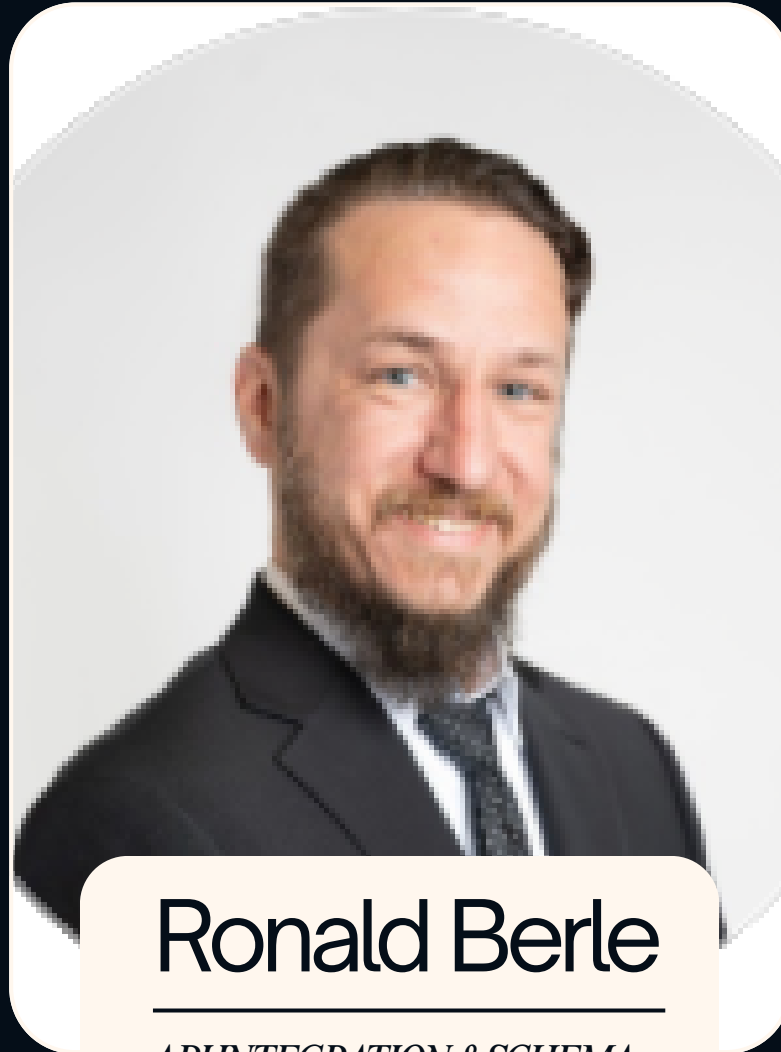
Amer Amer

FRONTEND & DATA OPERATIONS



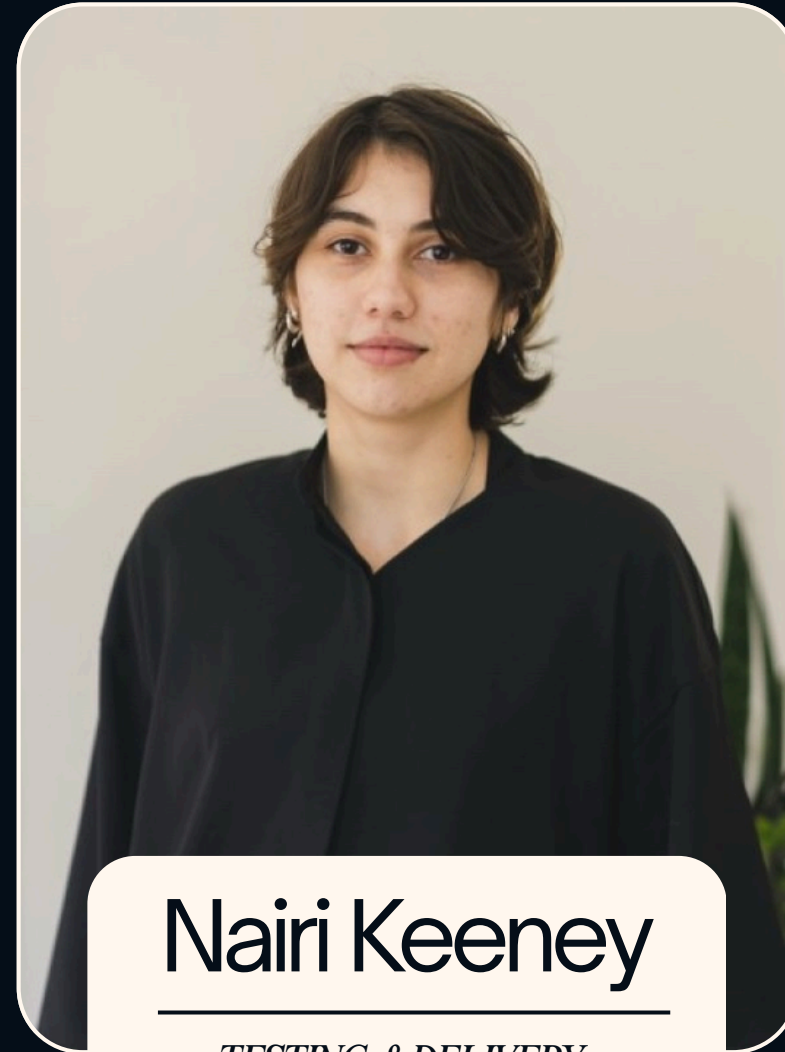
Nada Belafqih

AZURE DEPLOYMENT & DEVOPS



Ronald Berle

*API INTEGRATION & SCHEMA
DESIGN*



Nairi Keeney

TESTING & DELIVERY

Our Team //

A four-person team of USF graduate students combining expertise in frontend development, cloud deployment, API integration, and quality assurance to build a full-stack fitness analytics application.

Technical Intent //

Architecture (MVC)

- *Model: C# models*
- *View: HTML templates per entity*
- *Controller: Handles HTTP routes & logic*
- *Clean Separation of concerns throughout*

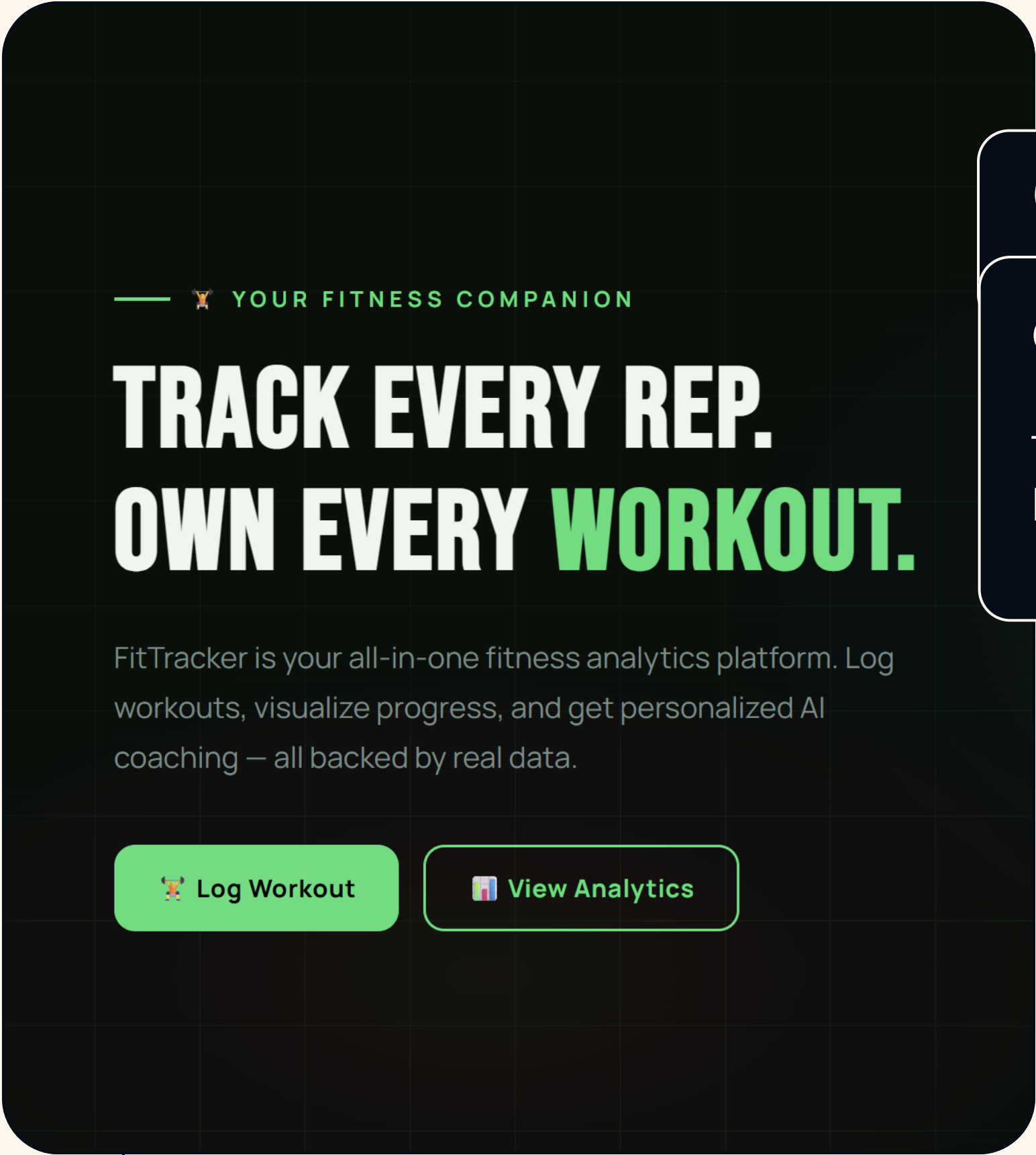
Technical Focus

- *Azure App Service deployment*
- *Persistent Local Storage*
- *REST API for exercise/nutrition data*
- *Responsive, mobile-friendly UI*

Goals

- *Track workouts, calories & health metrics*
- *Full CRUD for users, workouts & exercises*
- *Fetch live fitness data via external API*
- *Persis data across sessions*

Demo



CRUD Operations

CREATE

Add workouts, log exercises & save meal entries

READ

View dashboard stats, workout history & progress charts



UPDATE

Edit workout plans, update reps/sets & calories

DELETE

Remove entries, clear sessions & reset goals

Learning

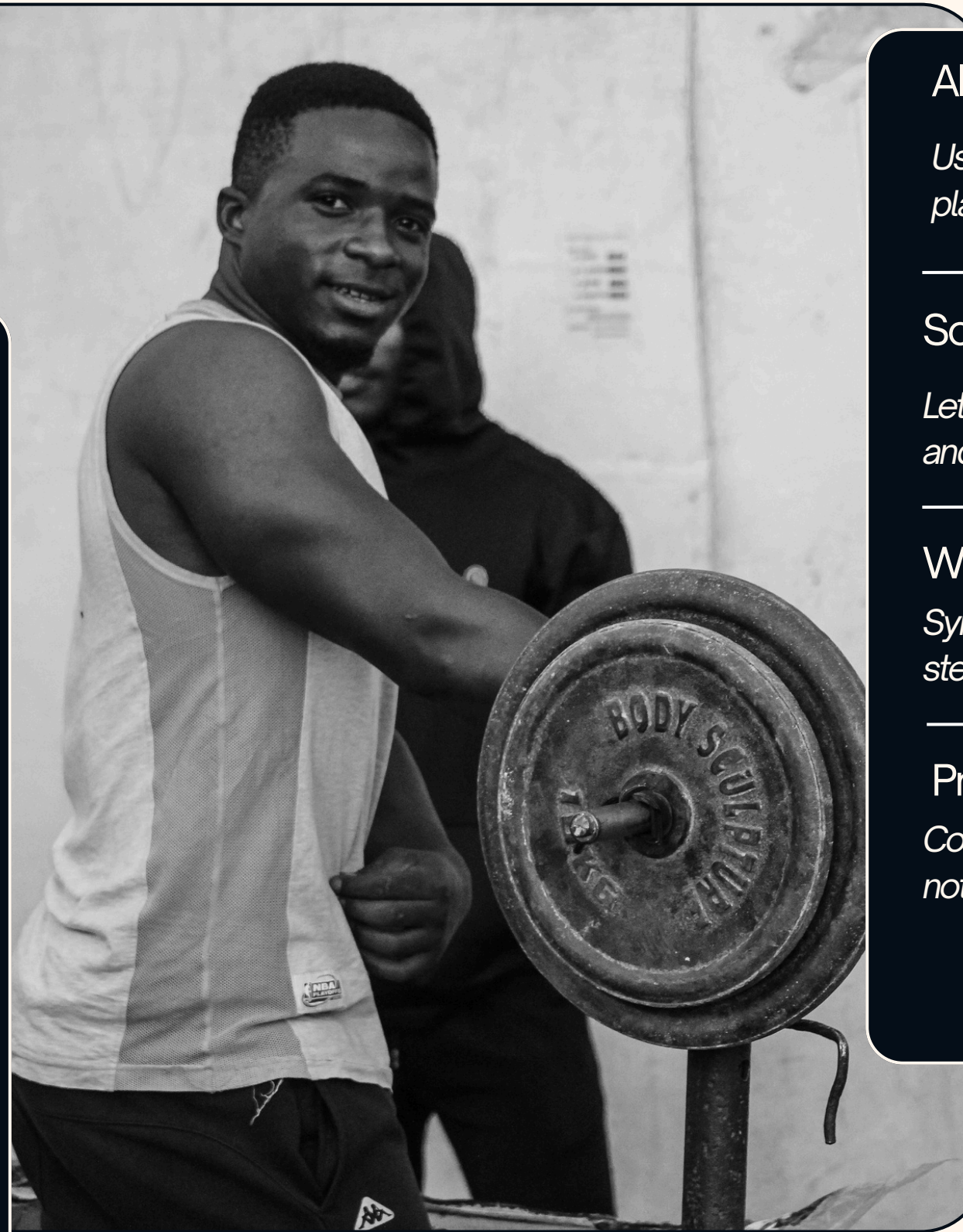
Key Learnings & Challenges

*MVC pattern enforces clean separation.
Debugging became simpler once controllers
were isolated from views*

*Designing ERD upfront saved significant rework
when relations between Workout and Exercise
entities grew complex*

*Azure deployment revealed environment-
specific config issues not caught locally.
Learned value of staging slots*

*Rate-limiting on fitness API required
implementing caching to avoid exceeding
free-tier quotas during testing*



AI Workout Planner

*Use Claude/GPT to generate personalized weekly
plans based on user history and goals*

Social Challenges

*Let users create group challenges, share progress,
and compare leadership boards with friends*

Wearable Integration

*Sync with Fitbit/Apple Health APIs to auto-import
steps, heart rate, and sleep data*

Progressive Web App

*Convert to a PWA for offline access, push
notifications, and home screen installation*

Ideas for Extension

& Extension



THANK YOU // *Any Questions?*